**Results**

The experiments compared four different prompting strategies across sentiment analysis and question answering. The prompting strategies evaluated were zero-shot prompting, few-shot prompting, chain-of-thought (CoT) prompting, and CoT self-consistency prompting. The experiments used the model meta-llama/Llama-3.1-8B-Instruct. All experiments were performed with a constant temperature of 0.2 and a maximum token limit of 200 for every model call. Five self-consistency thoughts were generated per example. For sentiment analysis, the model classified movie reviews as positive, negative, or neutral. The dataset was obtained from the IMDb dataset. For question answering (QA), the model answered factual questions using context paragraphs from the SQuAD dataset. Both IMDb and SQuaAD were downloaded from Hugging Face and used to build the dataset. 10 test inputs were randomly selected from each set. The outputs were evaluated using accuracy to the ground truth. Latency and token usage were measured for each prompting strategy. The number of prompt tokens, response tokens, and total tokens were recorded for each example. The results of the experiments can be seen below in Table 1., Figure 1, and Figure 2.

| Task | Technique | N examples | Accuracy | Avg Latency (s) | Avg total tokens | Avg prompt tokens | Avg response tokens |
|---|---|---|---|---|---|---|---|
| sentiment_analysis | zero_shot | 10 | 1 | 0.132 | 302.3 | 300.3 | 2 |
| sentiment_analysis | few_shot | 10 | 0.9 | 0.101 | 363.8 | 361.8 | 2 |
| sentiment_analysis | chain_of_thought | 10 | 1 | 2.684 | 506.8 | 306.8 | 200 |
| sentiment_analysis | self_consistency | 10 | 0.9 | 13.367 | 2504 | 1504 | 1000 |
| question_answering | zero_shot | 10 | 0.9 | 0.221 | 237.9 | 226.3 | 11.6 |
| question_answering | few_shot | 10 | 0.8 | 0.201 | 307.2 | 297.3 | 9.9 |
| question_answering | chain_of_thought | 10 | 1 | 1.850 | 377.9 | 241.3 | 136.6 |
| question_answering | self_consistency | 10 | 0.9 | 7.980 | 1718.7 | 1131.5 | 587.2 |

Table 1.

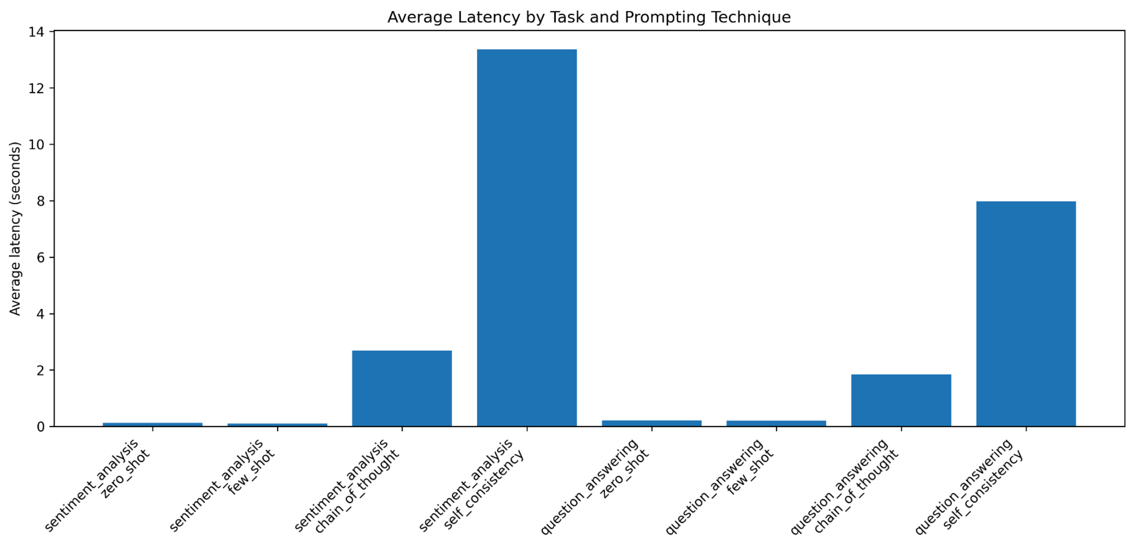


Figure 1.

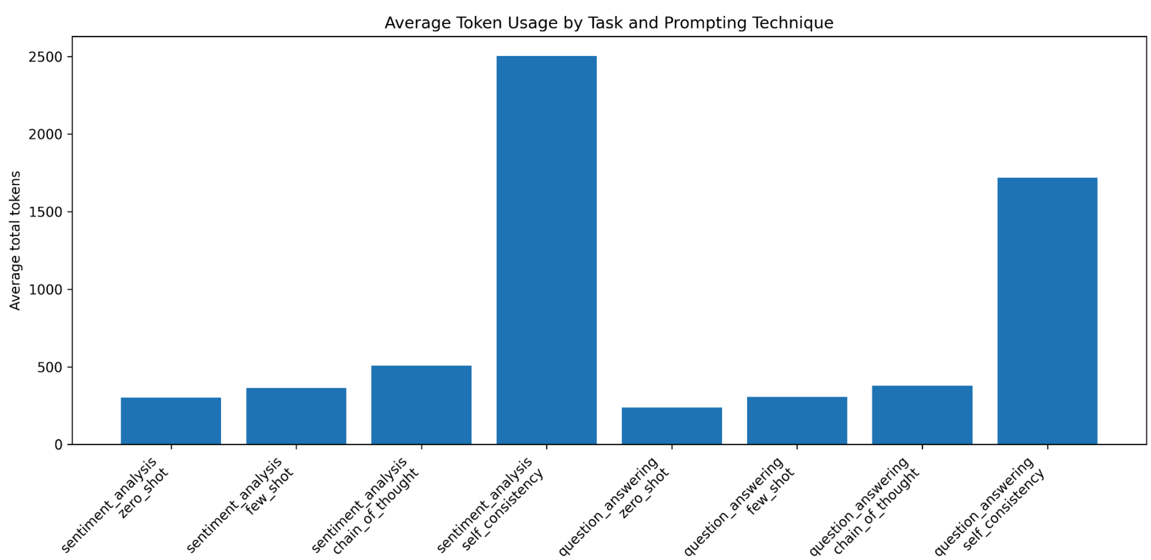


Figure 2.

For the results, they were consistent across both tasks. CoT prompting performed best overall as it had 100% accuracy on both sentiment analysis and QA. This suggests that instructing the model to reason step by step improved its ability to produce correct answers. Zero-shot prompting performed well as it achieved 100% accuracy on sentiment analysis.. This indicates that current models are capable at simple classification tasks without requiring examples. Few-shot prompting tended to decrease accuracy. A possible explanation is that the few-shot examples may not have been representative enough of the test data or the examples may have confused the model. Self-consistency also did not significantly improve results in this experiment.

### Cost/latency Tradeoff

Zero-shot prompting had the lowest latency and token cost because the prompts were short. Few-shot prompting increased prompt size as every request had multiple examples. This increased prompt token usage but did not significantly increase response tokens. CoT prompting significantly increased response tokens because of the intermediate reasoning steps before answering. Self-consistency had the highest cost for both latency and tokens due to the five separate model calls for every input. Sentiment analysis self-consistency averaged over 2500 total tokens per example, compared to only about 300 tokens for zero-shot prompting. Similarly, latency increased from approximately 0.1 seconds for zero-shot prompting to around 13 seconds for self-consistency. From the results of this experiment, it appears self- consistency did not outperform zero-shot prompting, especially when considering the lower accuracy. This may be due to the simple classification examples.

### Error Analysis

Few-shot prompting tended to produce worse results than zero-shot prompting. An example can be seen in sentiment analysis example 8 where the model labeled it as negative but was actually positive. The example contained a mixture of sentiments such as “excellent film” but also “painfully slow and the plot made no sense” and “it was average”. The model seemed to struggle with mixed or more nuanced sentiments. To improve this, the prompt could explicitly say to “classify based on the overall final opinion and not on specific phrases”. Another example was in QA example 7 where the response was “22,000” but the true answer was “over 22,000”. While it may have been mostly correct, it had slightly different wording or meaning, and thus may have required more accurate phrasing. To improve this error, the prompt could say “do not paraphrase or simplify numbers”.

### Lessons learned

A surprising result was how well zero-shot prompting performed compared to more complicated prompting systems. Its accuracy and low costs indicate that current models are highly capable without requiring examples for many tasks. Another surprising result was how much more significantly expensive self-consistency prompting is due to its many calls. For simple tasks such as analysing sentiment and answering simple questions, zero-shot prompting is very useful. For future prompt engineering work, using larger and more complicated datasets can provide more failure examples to improve the prompts. Most prompting strategies missed one out of 10 questions so may not give an accurate assessment of when they fail.